AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

APRIL/ MAY 2018

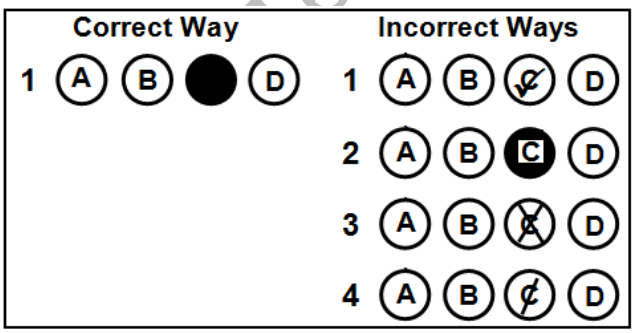
Mathematics Paper I

Time: 40 minutes Marks: 30

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. $\frac{b+c}{2}-\frac{b+c}{3}$ is equal to

   A. $-\frac{b+c}{6}$

   B. $-\frac{b+5c}{6}$

   C. $\frac{b+c}{6}$

   D. $\frac{b+5c}{6}$

2. The square root of $(x+2)^2\left(x^2-2x+1\right)$ is

   A. $\pm(x+2)(x-1)$

   B. $\pm(x+2)(x-2)$

   C. $\pm(x+1)(x-2)$

   D. $\pm(x+2)\left(x-\sqrt{2x}+1\right)$

3. The highest common factor (H.C.F) of $(x+y)^2$ and x^2+xy is

   A. $(x+y)^2$

   B. x^2+xy

   C. $x+y$

   D. $x(x+y)^2$

4. Which of the following fractions is a proper fraction?

   (**Note:** $x \neq a$ and b)

   A. $\frac{(x-b)^4}{(x-a)^2(x-b)}$

   B. $\frac{(x-b)^3}{(x-a)^2(x-b)}$

   C. $\frac{(x-a)^3}{(x-a)^2(x-b)}$

   D. $\frac{(x-a)^2}{(x-a)^2(x-b)}$

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5. The half of a number x increased by 3 is $\frac{12}{3}$. The equation representing the given situation is

A. $x^{\frac{1}{2}} - 3 = \frac{12}{3}$

B. $x^{\frac{1}{2}} + 3 = \frac{12}{3}$

C. $\frac{x}{2} - 3 = \frac{12}{3}$

D. $\frac{x}{2} + 3 = \frac{12}{3}$

6. If $2(x-b) = 0,$ then x will be equal to

A. $2 + b$

B. $2b$

C. b

D. $\frac{b}{2}$

7. The solution set of $\frac{\sqrt{x}}{-2} = -1$ is

A. ϕ

B. $\{2\}$

C. $\{4\}$

D. $\{-4\}$

8. The set of values satisfying $2 - x > 0,$ where $x \in W,$ is

A. $\{1, 2\}$

B. $\{0, 1\}$

C. $\{-1, -2\}$

D. $\{3, 4, 5, ...\}$

9. For linear equation $ax + by = 0$, if $y = a$, then x is equal to

(**Note**: a and b are constants)

A. $-a$

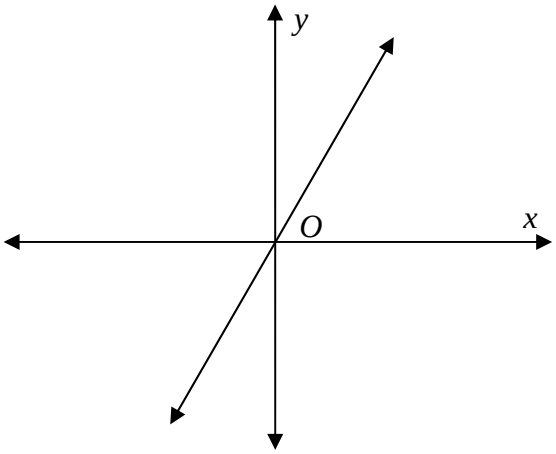
B. $-\frac{b}{a}$

C. $-b$

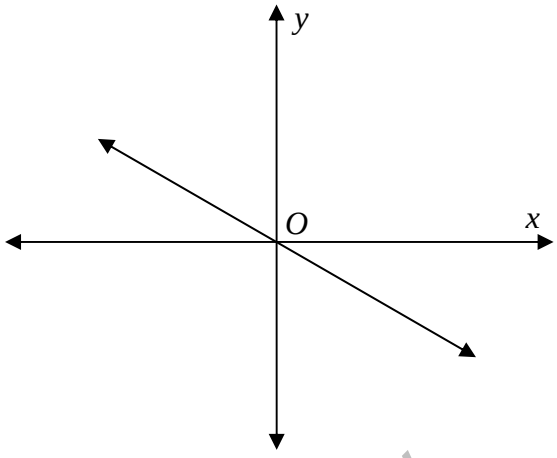
D. $-\frac{a^2}{b}$

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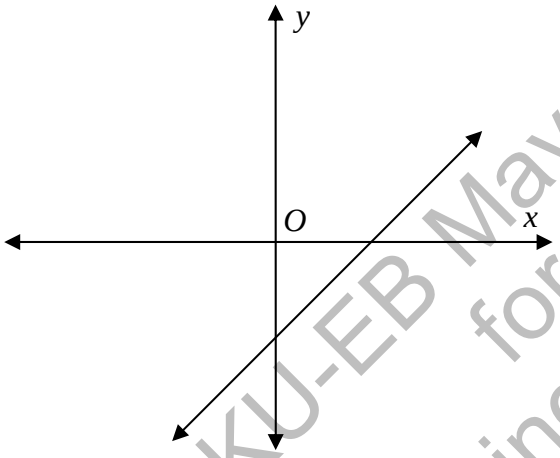
10. The CORRECT representation for the graph of $y = -3x - 3$ is



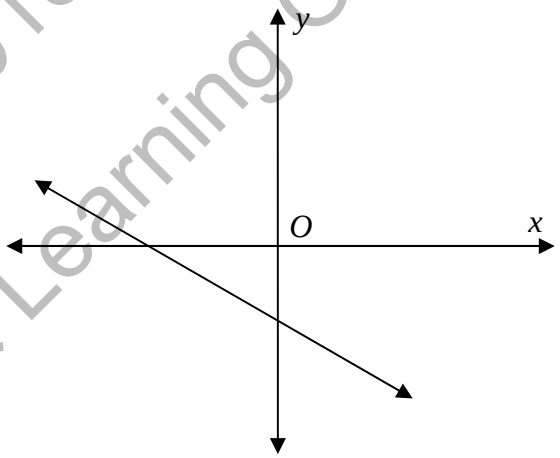
A



B



C



D

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11. If (a, b) is a point in the Cartesian plane and $a > 0$ and $b < 0$, then the point lies in the

A. I quadrant.
B. II quadrant.
C. III quadrant.
D. IV quadrant.

12. The solution set of quadratic equation $a^2x^2 = b^2$, where $x \in R$ and $a \neq 0$, is

A. $\left\{\frac{b}{a}\right\}$

B. $\left\{\pm\frac{b}{a}\right\}$

C. $\left\{\sqrt{b^2 - a^2}\right\}$

D. $\left\{\pm\sqrt{b^2 - a^2}\right\}$

13. Which of the following equations is a quadratic equation?

A. $ax + by + c = 0$

B. $ax + b = cx^2$

C. $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2ab}$

D. $c = \sqrt{(ax + b)^2}$

14. The solution set of $3x^2 = 1$, where $x \in R$, is

A. $\left\{\frac{1}{3}\right\}$

B. $\left\{\pm\frac{1}{3}\right\}$

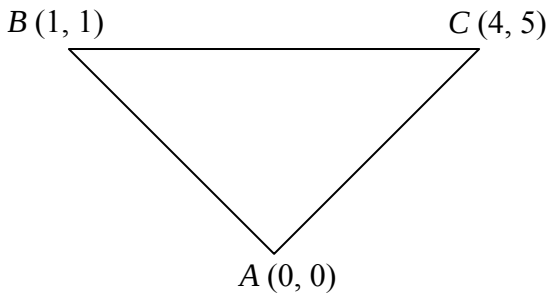
C. $\left\{\frac{1}{\sqrt{3}}\right\}$

D. $\left\{\pm\frac{1}{\sqrt{3}}\right\}$

15. In the given triangle *ABC*, side *BC* is equal to

NOT TO SCALE

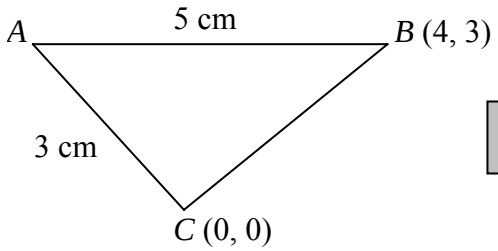
A. 5 units
B. 7 units
C. $\sqrt{41}$ units
D. $\sqrt{61}$ units



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16. In the given diagram, the triangle *ABC* is a/ an

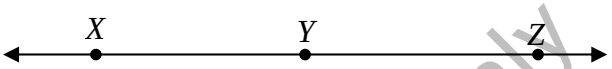
I. scalene triangle.
II. isosceles triangle.
III. right angled triangle.



NOT TO SCALE

A. I only
B. II only
C. I and III
D. II and III

17. If *X*, *Y* and *Z* are collinear points as shown in the figure, then which of the following conditions is correct?



A. $|XZ| = |XY| + |YZ|$
B. $|XY| = |XZ| + |YZ|$
C. $|XY| = |YZ| - |XY|$
D. $|XZ| = |YZ| - |XY|$

18. The angle $20^{o}30'$ expressed in decimal form as

A. 20.05
B. 20.50
C. 20.03
D. 20.30

19. The central angle subtended by an arc of length 2 m, at the centre of a circle whose diameter is 2 m, will be

A. 0.5°
B. 2°
C. 0.5 radians
D. 2.0 radians

20. $\dfrac{\sin^2\theta}{1-\sin^2\theta}$ is equal to

A. $-\tan^2\theta$
B. $\cot^2\theta$
C. $-\cot^2\theta$
D. $\tan^2\theta$

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21. $\sqrt{2}\cos 45°$ is equal to

A. $\frac{1}{2}$
B. $\frac{1}{\sqrt{2}}$
C. 1
D. 2

22. The lengths of the sides of a rectangular table are 12 feet and 5 feet. The length of its diagonal will be

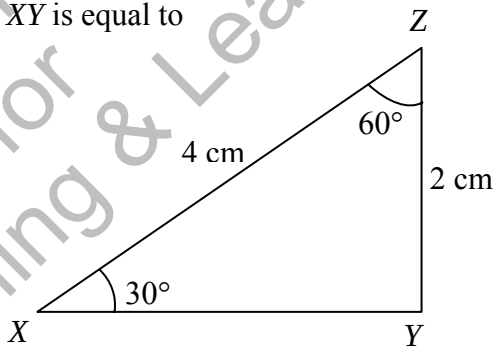
A. 10.9 feet
B. 12.5 feet
C. 13.0 feet
D. 17.0 feet

23. The possible measurements of the sides of a right angled triangle are

A. 1 m, 1 m, and 2 m
B. 3 m, 4 m, and 5 m
C. 5 m, 7 m, and 9 m
D. 2 m, 3 m, and 4 m

24. In the given triangle XYZ, side XY is equal to

A. 2 cm
B. $2\sqrt{3}$ cm
C. 3 cm
D. 4 cm



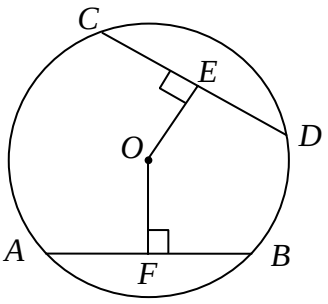
NOT TO SCALE

25. If (3, 0), (5, 0) and (0, 2) are three points, then the number of circle(s) passing through these points is

A. 0
B. 1
C. 3
D. infinite

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26. In the given diagram, O is the centre of circle. If AB = 8 cm and $OE = OF$, then CD will be equal to



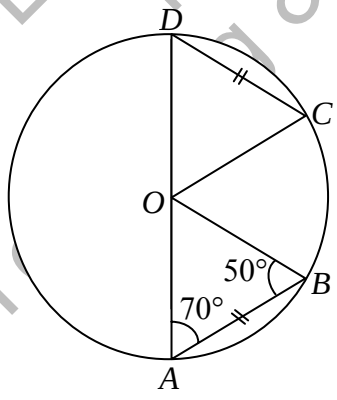
NOT TO SCALE

A. 3 cm
B. 4 cm
C. 5 cm
D. 8 cm

27. If two circles touch each other internally and their radii are c and b, where $c > b$, then distance between their centres is

A. $c - b$
B. $\frac{c-b}{2}$
C. $\frac{b+c}{2}$
D. $b + c$

28. In the given diagram, O is the centre of the circle and $\overline{AB} = \overline{CD}$. The $\angle DOC$ is equal to

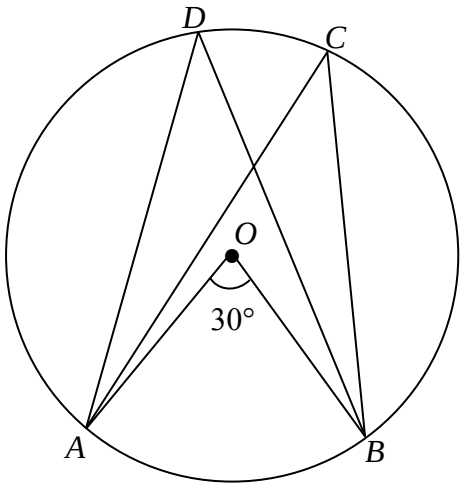


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A. 50°
B. 60°
C. 70°
D. 90°

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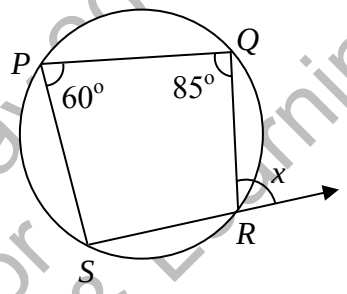
29. In the following diagram, a circle with centre O is given. The $\angle ACB + \angle ADB$ is equal to



NOT TO SCALE

A. 15°
B. 30°
C. 60°
D. 120°

30. In the given figure, the value of x is equal to



NOT TO SCALE

A. 60°
B. 85°
C. 95°
D. 120°

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